

SMART VENTILATION SYSTEM USING ARDUINO UNO

Components Required

Arduino Uno
MQ2 Gas Sensor Module
SG90 Servo Motor
Red LED
220 Ω Resistor
Breadboard
Jumper Wires
USB Cable

Working

The Smart Ventilation System uses an Arduino Uno with an MQ2 gas sensor to continuously monitor the concentration of LPG, smoke and other combustible gases. When the detected gas value exceeds the preset threshold, the Arduino turns ON a red LED and rotates an SG90 servo motor to 90 degrees to simulate opening a ventilation window. When the gas level returns to a safe value, the LED turns OFF and the servo returns to the closed position. This project is suitable for smart homes, kitchens, laboratories and industrial safety applications.

Connections

Component	Arduino Uno Pin
MQ2 VCC	5V
MQ2 GND	GND
MQ2 A0	A0
Servo Signal	D9
Servo VCC	5V
Servo GND	GND
LED Anode (+)	D7 (through 220 Ω resistor)
LED Cathode (-)	GND

PROJECT LINK

<https://wokwi.com/projects/469959241263001601>